

Software Design Document (SDD)

E-Commerce with mock payment developers

E-Commerce System with Mock Payment Integration

Software Design Document

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1. INTRODUCTION

1.1 Purpose

This Software Design Document (SDD) describes the architecture and system design of the E-Commerce System with Mock Payment Integration. The document is intended for developers, system analysts, testers, and project supervisors involved in the development and maintenance of the system.

The purpose of this system is to provide users with an online platform where they can browse products, manage shopping carts, place orders, and simulate online payments using a mock payment gateway.

1.2 Scope

The E-Commerce System is a web-based application developed to support online shopping activities. The system allows customers to:

- Register and log into accounts
- Browse products by category
- Add products to a shopping cart
- Place orders
- Simulate payments using a mock payment gateway
- Track order status

The system also provides an admin panel where administrators can:

- Manage products
- Manage categories
- View customer orders
- Update order statuses

Objectives

- Provide a simple and efficient online shopping experience
- Simulate secure payment processing for learning purposes
- Improve inventory and order management

Benefits

- Reduces manual sales processes
- Provides convenient shopping access
- Enhances product management efficiency
- Demonstrates payment workflow integration

1.3 Overview

This document provides detailed information about the design of the E-Commerce System. It includes system architecture, data design, component design, interface design, and requirement traceability.

1.4 Reference Material

The following materials were used during the preparation of this document:

- IEEE Std 1016 Software Design Description Guidelines
- HTML, CSS, JavaScript Documentation
- PHP Documentation
- MySQL Documentation
- Software Requirements Specification (SRS) for the E-Commerce System

1.5 Definitions and Acronyms

Term/Acronym	Meaning
SDD	Software Design Document
SRS	Software Requirements Specification
UI	User Interface
DBMS	Database Management System
API	Application Programming Interface
Mock Payment	Simulated payment processing system
Admin	System Administrator

2. SYSTEM OVERVIEW

The E-Commerce System is a web application that enables customers to purchase products online. Users can create accounts, browse available products, add items to a cart, and simulate payment transactions.

The system consists of three major modules:

1. Customer Module
2. Admin Module
3. Payment Simulation Module

The mock payment integration simulates real-world payment processing without using actual banking systems.

3. SYSTEM ARCHITECTURE

3.1 Architectural Design

The system uses a three-tier architecture:

Presentation Layer

Handles user interaction through web pages.

Application Layer

Processes business logic such as:

- Authentication
- Cart management
- Order processing
- Payment simulation

Database Layer

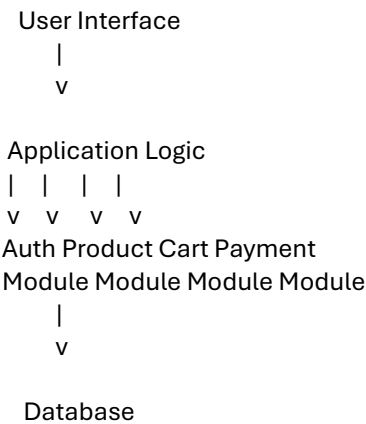
Stores:

- User data
- Product information
- Orders
- Payment records

Major Subsystems

Subsystem	Responsibility
Authentication Module	Handles registration and login
Product Management Module	Manages products and categories
Shopping Cart Module	Handles cart operations
Order Management Module	Processes customer orders
Mock Payment Module	Simulates payment processing
Admin Dashboard	Allows admin control and monitoring

System Architecture Diagram



3.2 Decomposition Description

Functional Modules

1. Authentication Module

Functions:

- Register User
- Login User
- Logout User

2. Product Module

Functions:

- Add Product
- Update Product
- Delete Product
- Search Product

3. Shopping Cart Module

Functions:

- Add to Cart
- Remove from Cart
- Update Quantity

4. Order Module

Functions:

- Create Order
- View Orders
- Update Order Status

5. Mock Payment Module

Functions:

- Validate Payment Details
- Simulate Transaction
- Generate Payment Confirmation

Sequence Diagram (Order Process)

Customer -> Cart -> Checkout -> Mock Payment -> Order Confirmation

3.3 Design Rationale

The three-tier architecture was selected because it:

- Separates responsibilities
- Simplifies maintenance
- Improves scalability
- Enhances security

Alternative architectures such as monolithic design were considered but rejected because they reduce maintainability and flexibility.

The mock payment gateway was chosen to safely simulate online transactions without requiring integration with real banking APIs.

4. DATA DESIGN

4.1 Data Description

The system uses a relational database to manage application data.

Main Entities

- Users
- Products
- Categories
- Cart
- Orders
- Payments

Database Tables

Table Name	Description
users	Stores customer and admin information
products	Stores product details
categories	Stores product categories

Table Name	Description
Cart	Stores temporary shopping cart items
orders	Stores customer orders
payments	Stores payment transaction records

4.2 Data Dictionary

Users Table

Field Name	Type	Description
user_id	INT	Unique user identifier
full_name	VARCHAR	Customer full name
email	VARCHAR	User email
password	VARCHAR	Encrypted password
Role	VARCHAR	User role

Products Table

Field Name	Type	Description
Product id	INT	Unique product ID
Product name	VARCHAR	Product name
description	TEXT	Product description
price	DECIMAL	Product price
Stock quantity	INT	Available stock

Orders Table

Field Name	Type	Description
Order id	INT	Unique order ID
User id	INT	Customer ID
Total amount	DECIMAL	Total order cost
Order status	VARCHAR	Current order status
Order date	DATE	Date of order

5. COMPONENT DESIGN

Authentication Module

Login Function Pseudocode

```
BEGIN
  INPUT email and password
  VALIDATE credentials
  IF credentials are correct THEN
    GRANT access
  ELSE
    DISPLAY error message
  ENDIF
END
```

Shopping Cart Module

Add to Cart Function

```
BEGIN
  SELECT product
  CHECK product availability
  ADD product to cart
  UPDATE cart total
END
```

Mock Payment Module

Payment Processing Function

```
BEGIN
  INPUT mock card details
  VALIDATE details
  IF valid THEN
    DISPLAY payment success
    GENERATE transaction ID
  ELSE
    DISPLAY payment failed
  ENDIF
END
```

6. HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

The system provides a user-friendly web interface where users can:

- Browse products
- Manage shopping carts
- Make mock payments
- View orders

Admins can:

- Manage inventory
- Monitor orders
- Manage users

6.2 Screen Images

Main Screens

1. Home Page
2. Login/Register Page
3. Product Listing Page
4. Shopping Cart Page
5. Checkout Page
6. Mock Payment Page
7. Admin Dashboard

Example Wireframe

Logo	Search Bar	Cart
Categories		
Product Image		
Product Name		
Price		
Add to Cart		

6.3 Screen Objects and Actions

Screen Object	Action
Login Button	Authenticates user
Add to Cart Button	Adds product to cart
Checkout Button	Starts payment process
Pay Now Button	Simulates payment
Admin Product Form	Adds or edits products

7. REQUIREMENTS MATRIX

Requirement ID	Requirement Description	Component
FR-01	User registration	Authentication Module
FR-02	User login	Authentication Module
FR-03	Browse products	Product Module
FR-04	Add items to cart	Shopping Cart Module
FR-05	Process mock payment	Payment Module
FR-06	Generate orders	Order Module
FR-07	Admin manages products	Admin Dashboard

8. APPENDICES

Appendix A: Suggested Technologies

Technology	Purpose
HTML/CSS	Frontend Design
JavaScript	Client-side functionality
PHP	Backend development
MySQL	Database management
XAMPP	Local server environment

Appendix B: Future Improvements

- Real payment gateway integration
- Mobile application support
- Product reviews and ratings
- Email notifications
- Delivery tracking system